Sensor Dashboard Assessment – Procedure and Tools Used

Project Overview

The objective of this project was to develop a simple Sensor Dashboard that displays dummy sensor data and provides an API endpoint to retrieve and update the data. The application was developed using Flask for the backend and HTML, CSS, and JavaScript for the frontend.

Features Implemented

- Display Temperature, Humidity, and Pressure values.
- Simple dashboard interface.
- REST API endpoint for sensor data.
- GET operation to retrieve data.
- POST operation to update data.
- Frontend and backend integration.

Technologies Used

Backend: Python, Flask
Frontend: HTML, CSS, JavaScript
Tools: VS Code, Git, GitHub, Postman, Google Chrome

Implementation Procedure

1. Developed a Flask backend application.
2. Created a REST API endpoint (/sensor).
3. Built a dashboard using HTML, CSS, and JavaScript.
4. Connected frontend and backend using Fetch API.
5. Tested GET and POST requests using Postman.
6. Uploaded source code to GitHub.

Outcome

The Sensor Dashboard was successfully developed and tested. The project demonstrates backend development, API implementation, frontend-backend communication, and version control using GitHub.

GitHub Repository: https://github.com/madhesh3905/Sensor-Dashboard-Assessment

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